Guided Feature Fusion: Zero-Shot Framework for Hyperspectral Image Denoising

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***Abstract*—Hyperspectral image (HSI) denoising algorithms often face challenges with nonuniform, band-dependent noise and fail to fully utilize high signal-to-noise ratio (SNR) spectral bands. To address these issues, this article presents a zero-shot framework that leverages high-SNR bands as scene-specific references to guide the restoration of degraded bands. The framework features a dual-branch network architecture with the guide branch extracting high-fidelity spatial structures from high-SNR bands and the denoising branch restoring degraded bands through dynamic feature fusion. Fusion modules, enhanced with spectral–spatial attention, enable adaptive and spatially aware refinement between branches. Unlike conventional methods relying on external training datasets or fixed priors, the proposed approach adapts to the scene-specific quality disparities within each HSI. Extensive experiments on synthetic and real-world datasets demonstrate its superior ability in preserving spatial–spectral fidelity and handling complex noise scenarios, outperforming state-of-the-art methods.**

***Index Terms*—Hyperspectral image denoising (HSI), mixture noise removal, unsupervised deep learning.**

I. Introduction

HYPERSPECTRAL imaging captures data across contiguous spectral bands, providing rich spectral information unattainable with traditional color imaging. This high spectral resolution enhances the ability to identify and classify objects in complex scenes, making hyperspectral image (HSI) an indispensable tool for applications such as remote sensing, environmental monitoring, and precision agriculture [1], [2], [3], [4]. However, HSI data acquisition is often compromised by various types of noise, including Gaussian noise, stripe noise, and impulse noise [5], [6], which result from sensor limitations, calibration errors, and environmental factors. These noise artifacts severely degrade the quality of HSI data, negatively affecting the performance of critical analysis tasks, such as terrain classification and object detection [7], [8], [9]. Consequently, effective denoising of HSI has become an essential preprocessing step to preserve data integrity and ensure accurate and reliable analysis outcomes.

Existing methods for HSI denoising can be divided into two main categories: deep learning methods and model-based approaches. Model-based approaches rely on statistical priors, such as low-rankness, nonlocal self-similarity (NSS), and local smoothness, to regularize the denoising process. Low-rankness captures spectral correlations by compressing the high-dimensional spectral features of HSI into a low-rank subspace, effectively suppressing noise [10]. NSS leverages the presence of similar patches across the entire HSI, sharing information to reduce noise and enhance structural details [11]. Local smoothness exploits the similarity of neighboring pixel values, using techniques like total variation (TV) regularization to smooth noise while preserving crucial edge features [12], [13]. These model-based methods typically offer strong generalization to unseen noise types and produce more interpretable results compared to other approaches.

Deep learning-based methods involve training deep neural networks on paired noisy and clean datasets, enabling the network to learn a direct mapping from noisy inputs to clean outputs [14], [15]. For instance, Chang et al. [16] proposed an HSI denoising method using multichannel filters, while Yuan et al. [17] utilized multiscale feature extraction and representation to enhance the denoising effect. Additionally, Shi et al. [18] developed a joint spectral–spatial denoising strategy leveraging a 3-D attention mechanism to capture spectral–spatial correlations within HSIs. While these methods have demonstrated remarkable denoising capabilities, their performance heavily depends on the availability and quality of training data. Insufficient or unrepresentative training datasets can significantly degrade the model's effectiveness. Moreover, the high-computational cost of acquiring hyperspectral data and the requirement for large annotated datasets pose substantial challenges for practical applications.

A critical limitation of many existing methods is their tendency to treat all spectral bands equally during denoising, without accounting for the inherent variations in noise levels across spectral bands. In practice, hyperspectral datasets often exhibit significant disparities in noise intensity, with some bands suffering from severe noise and others maintaining a high signal-to-noise ratio (SNR). For instance, in the EO-1 [19] and GAOFEN-5 datasets [20] (detailed in Section III),

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certain bands are heavily degraded by noise (e.g., Gaussian and stripe noise), while others retain clear spatial structures. High-SNR bands are particularly important as they preserve essential information that can guide the restoration of noisier bands. This observation highlights the need for a more adaptive and flexible denoising approach that dynamically adjusts to the noise characteristics of each spectral band. By leveraging the high-quality information in high-SNR bands, such an approach can significantly enhance overall denoising performance.

To address these challenges, we propose a novel zero-shot framework that diverges from conventional approaches. Our method is inspired by a key observation in real-world hyperspectral imagery: within the same hyperspectral cube, certain bands retain exceptional clarity and structural fidelity, while others are significantly degraded. These high-SNR bands serve as natural, scene-specific references, preserving authentic spatial structures and details that can guide the restoration of degraded bands, thereby eliminating the need for external reference data or predefined priors.

The framework is built on two essential insights: 1) high-fidelity spatial information preserved in high-SNR bands provides reliable, scene-specific structural references that remain uncontaminated by severe noise; and 2) these clear bands can offer direct guidance for restoring degraded counterparts, enabling a more adaptive and scene-specific restoration approach compared to traditional methods that depend on external training data or fixed mathematical priors. This direct guidance strategy allows for the establishment of a data-specific restoration mechanism that dynamically adapts to each scene's unique characteristics.

Our framework integrates these principles through an innovative dual-branch architecture explicitly designed to exploit the quality disparity between bands. The guide branch incorporates hierarchical feature extraction mechanisms tailored to leverage high-quality spatial structures from clear bands. It captures both fine-grained details and global patterns that characterize the authentic properties of the scene. This branch employs cascaded convolutional layers with adaptive normalization strategies to ensure robust feature extraction across varying imaging conditions.

The interaction between branches is facilitated through our proposed quality-guided fusion modules, which implement an advanced cross-band information transfer mechanism. These modules enhance the framework in several complementary ways: 1) quality-aware feature alignment adapts to the unique characteristics of each scene to ensure precise correspondence between high-SNR and degraded bands; 2) dynamic attention mechanisms selectively utilize high-fidelity spatial features from clear bands to prioritize the most informative regions; and 3) multiscale fusion strategies provide comprehensive structural guidance at various levels of detail to ensure both global consistency and fine-grained accuracy. This integrated design enables the framework to effectively restore degraded bands while preserving the authenticity of the original scene's characteristics. The key contributions of this article are summarized as follows.

1) Proposed a zero-shot framework that addresses the uneven noise distribution in real-world HSI data. By leveraging high-SNR bands to guide the restoration of low-SNR bands, the method eliminates the need for clean training data.
2) Designed a dual-branch network with spectral–spatial attention unit (SSAU)-based fusion modules, integrating guiding feature enhancement module (GFEM) and upsampling fusion enhancement module (UFEM) to enhance denoising performance by leveraging high-SNR features while preserving spatial–spectral details.
3) Demonstrated strong denoising performance and generalization through extensive experiments on simulated and real-world datasets, exhibiting advantages over state-of-the-art methods on both simulated and real-world data, all without requiring training.

The rest of this article is organized as follows. Section II provides an overview of related work in HSI denoising. Section III describes the proposed denoising network and its components in detail. Section IV presents experimental results along with a discussion of the findings. Finally, Section V concludes with a summary of key insights and directions for future research.

II. Related Work

This section reviews prevailing methods for HSI denoising, categorized into two primary approaches: traditional model-based methods and data-driven deep learning techniques. In addition, we also discuss related studies on HSI noise estimation.

A. Model-Based Approaches

Traditional model-based approaches to HSI denoising rely on utilizing prior knowledge about the intrinsic properties of HSIs. These methods typically frame denoising as an inverse problem, using statistical priors to regularize the solution space. Commonly used priors include low-rank structures, NSS, and TV regularization.

Low-rank methods assume that HSI data resides in a low-dimensional subspace [21], exploiting correlations among spectral bands. Techniques such as nuclear norm minimization [22] and LRTDTV [23] incorporate low-rank regularization to capture global spectral correlations while preserving spatial structures. Recent tensor-based approaches, such as Tucker decomposition [24] and tensor ring decomposition [25], further enhance the preservation of spectral–spatial structures by operating directly on the 3-D HSI data, leading to improved denoising results.

NSS exploits the inherent redundancy in HSIs by leveraging similar patterns across spatial locations to improve restoration accuracy [26]. Approaches like tensor dictionary learning (TDL) [11] and Kronecker-basis-representation (KBR) [27] utilize this property to model correlations between spatial patches. While effective in reducing structured noise, these methods often suffer from high-computational complexity and are less robust to sparse, unstructured noise, limiting their practical use in real-world applications.

TV regularization imposes smoothness constraints on the spatial and spectral gradients of HSIs [28], [29], [30]. Methods

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such as LLRSSTV [31] and enhanced 3D-TV (E3DTV) [32] are designed to maintain spatial smoothness while preserving spectral fidelity. These approaches are particularly effective in handling mixed noise by exploiting gradient sparsity across spectral bands. However, a key limitation of TV-based methods is their tendency to oversmooth fine spatial details, particularly in high-SNR bands, which can result in suboptimal denoising performance.

B. Deep Learning-Based Methods

Deep learning techniques have proven highly effective for HSI denoising, enabling the learning of direct mappings from noisy to clean images without relying on explicit physical models [33]. Early approaches primarily utilized 2-D convolutional neural networks (2D CNNs) to extract spatial features within individual spectral bands [34]. For instance, Chang et al. [16] used 2D CNNs to capture spatial structures but failed to fully exploit the rich spectral information inherent in HSIs.

To address the limitations of 2D CNNs in capturing spectral correlations, subsequent methods incorporated 3-D convolutional neural networks (3D CNNs) [35], such as 3-D U-Net [36] and QRNN3D [37]. These models directly process the HSI data cube, modeling both spectral and spatial correlations comprehensively. By leveraging spectral–spatial features, they achieved significant improvements in denoising performance. However, their complex architectures often function as "black boxes," raising concerns about interpretability and limiting the integration of domain-specific knowledge.

To improve interpretability and incorporate prior knowledge about HSIs, hybrid methods that combine model-based principles with deep learning frameworks have been proposed. Approaches such as T3SC [33] and SMDS-Net [15] integrate sparse coding and subspace modeling into deep networks, effectively unrolling iterative algorithms within a deep learning context. While these hybrid models enhance both denoising performance and interpretability, they typically rely on large amounts of paired noisy-clean training data. Additionally, their dependence on sensor-specific characteristics can limit their generalizability across different sensors and noise conditions.

A major challenge for existing deep learning-based methods is their reliance on extensive paired datasets, which are difficult to obtain in practice due to the scarcity of clean HSIs [38]. Real-world HSIs often exhibit complex and variable noise patterns, making it challenging for models trained on specific noise types to generalize effectively. Moreover, such models are prone to overfitting, leading to reduced performance when applied to diverse scenarios.

There is also research focused on restoring noise-degraded images without requiring paired training data. Techniques such as deep image prior (DIP) [39], [40] leverage untrained convolutional neural networks to iteratively optimize restoration tasks. These methods accept randomly initialized noise as input and generate a restored image as output, effectively reducing Gaussian noise. Niresi and Chi [41] extended this approach to HSI restoration, demonstrating its potential for denoising tasks without relying on pre-collected training datasets. However, while these zero-shot methods show promise, they often fall short of state-of-the-art performance, particularly when addressing the diverse and complex noise types prevalent in HSIs [42]. These limitations highlight the need for a robust zero-shot denoising framework capable of leveraging the intrinsic properties of HSIs without relying on paired training data. Such a framework should exhibit strong generalization capabilities and robustness, effectively adapting to diverse noise environments while significantly improving denoising performance. This would address the shortcomings of existing methods and provide a more versatile solution for complex hyperspectral noise.

C. Hyperspectral Image Noise Estimation

Recent years have witnessed significant advances in HSI noise estimation. Statistical methods leverage intrinsic characteristics of hyperspectral data for noise intensity estimation. The HySime algorithm [43], for example, estimates noise by constructing a multiple regression model, treating each spectral band as a linear combination of other bands. Regression coefficients are calculated through the least squares method, and residuals from this regression are used as noise estimates. Li and Sun [44] partition the HSI into superpixels, apply linear regression to decouple signal and noise, and estimate the standard deviations of signal-independent and signal-dependent noise using maximum-likelihood optimization with conjugate gradient methods. Additionally, MixFus [45] selects clean spectral bands via optimal neighborhood reconstruction and employs Gaussian mixture models for noise estimation. RoSEGS [46] introduces an outlier detection strategy to effectively identify sparse noise.

Data-driven deep learning approaches have strong performance in both HSI noise estimation and denoising. Specifically, these methods embed the estimated noise information into the network architecture or feature maps, allowing it to be fully exploited during the denoising process and thereby effectively enhancing network performance. For example, Yuan et al. [47] proposed Partial-DNet, which combines noise estimation with a channel attention mechanism, adaptively incorporating estimated band-wise noise intensities into feature maps to effectively handle spatially varying mixed noise. Li et al. [48] incorporated Stein's unbiased risk estimator into the self-supervised learning phase to adaptively estimate noise levels for each degraded HSI, offering reliable noise priors for network training. Pan et al. [49] proposed Sparse-QRNN, which models the spectral noise distribution and yields precise band-wise noise-intensity estimates.

Unlike the above methods that strive for accurate noise intensity estimation, our framework requires only the relative ordering of noise intensity to distinguish high-SNR from low-SNR bands. This design improves the denoising framework's robustness to noise estimation errors while providing reliable guidance to the network, thus enhancing denoising performance.

III. Proposed Method

A. Motivation

In real-world hyperspectral datasets, noise distribution across spectral bands is highly uneven: many bands are heavily

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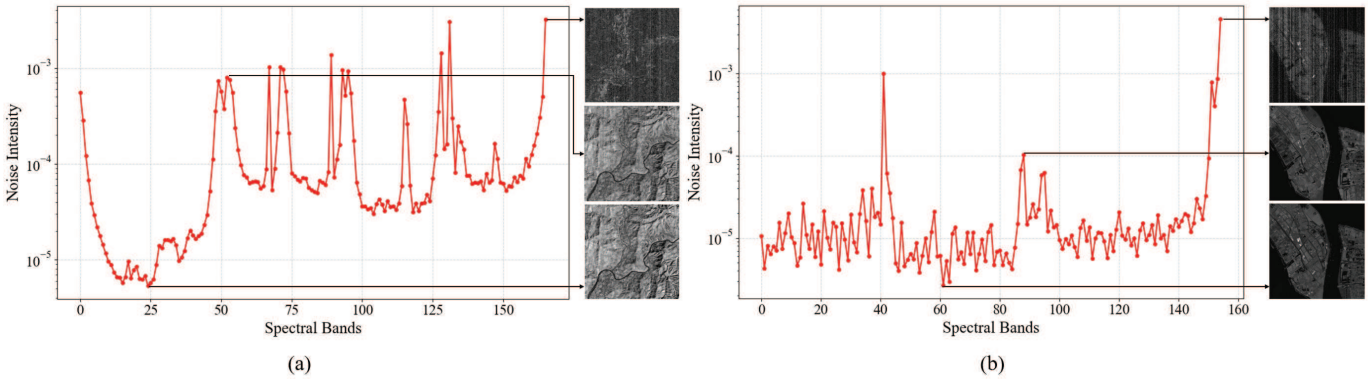


Fig. 1. Noise intensity distribution in real hyperspectral data. (a) EO-1. (b) GF-5.

affected by noise, while a few exhibit high SNR. To investigate this phenomenon, we analyzed two representative HSI datasets: EO-1 and GAOFEN-5 (GF-5). The EO-1 dataset, captured by the Hyperion sensor, spans wavelengths from 0.4 to 2.5 μm (400–2500 nm). Similarly, the GF-5 dataset, acquired using the Advanced Hyperspectral Imager on the GF-5 satellite, covers the same wavelength range [50].

To quantify noise levels, we applied the HySime algorithm to each spectral band and plotted the noise intensity distribution against the band index, as illustrated in Fig. 1. The resulting plots reveal significant variations in noise intensity and type across spectral bands in both datasets. Some bands are severely affected by noise, including Gaussian noise and stripe noise, while others retain intact spatial structure information, highlighting the uneven nature of noise distribution in real-world HSIs.

In the EO-1 dataset, bands in the mid-infrared and near-infrared regions (e.g., the 166th band) are particularly prone to higher noise levels, including significant stripe noise and deadline noise. This is likely due to the inherent sensitivity limitations of the Hyperion sensor's detectors at these wavelengths, resulting in reduced detection quality. Similarly, in the GF-5 dataset, severe stripe noise and deadline noise are observed in the 155th band, which may be attributed to detector defects or sensitivity limitations, as well as inconsistencies in hyperspectral sensor calibration across bands, particularly in longer wavelengths such as the infrared region.

Despite the rich prior information available in HSIs, many existing denoising methods fail to fully utilize this potential, especially the valuable data present in high-SNR bands. Using the HySime algorithm, we estimated the noise levels of each spectral band and introduced a guidance ratio to differentiate between high- and low-SNR bands. This ratio represents the proportion of bands classified as guiding bands based on a predefined noise threshold. Bands with noise intensities below the threshold were designated as guiding bands, as they contain high-quality spatial structural information and do not require restoration. The remaining bands, affected by higher noise levels, were classified as noisy bands in need of denoising. This dynamic selection process allows for the identification of high-quality bands within any HSI dataset, enabling a tailored denoising strategy.

Building upon these insights, we propose a novel denoising framework that effectively leverages the intrinsic prior information within HSI data. In our approach, high-SNR guiding bands provide crucial spatial structural priors, while low-SNR bands are enhanced using their own spectral prior information. To achieve this, we designed a network architecture that simultaneously learns from the high-quality spatial structures of guiding bands and integrates the inherent spectral correlations within hyperspectral data. By jointly leveraging spatial and spectral priors, our method enables high-fidelity reconstruction and denoising of noisy bands, effectively addressing the challenges posed by uneven noise distribution.

B. Framework

In real-world HSI denoising scenarios, the observed noisy HSI is represented as $\mathcal{Y} \in \mathbb{R}^{b\times h\times w}$, where b, h, and w denote the number of spectral bands, height, and width, respectively. HSIs are affected by various types of noise, which can be abstracted into an additive model

$$\mathcal{Y} = \mathcal{X} + \mathcal{N} \tag{1}$$

where $\mathcal{X}$ is the clean HSI and $\mathcal{N}$ represents the noise. The goal is to accurately recover the clean HSI $\mathcal{X}$ from the noisy observations $\mathcal{Y}$. The recovery of $\mathcal{X}$ from observed data $\mathcal{Y}$ poses significant challenges, given both noise interference and the inherent high-dimensional nature of HSIs. To improve the denoising process, a regularization term incorporating prior information about HSIs is introduced, leading to the following optimization problem:

$$\min_{\mathcal{X}} f(\mathcal{Y} - \mathcal{X}) + \lambda\phi(\mathcal{X}) \tag{2}$$

where f is a metric function measuring the difference between the noisy observations $\mathcal{Y}$ and the estimated clean HSI $\mathcal{X}$, $\phi(\mathcal{X})$ is a regularization term encoding prior knowledge, and λ is a regularization parameter. However, existing methods often fail to fully exploit the intrinsic relationships among spectral bands, especially when noise levels vary significantly across bands. To address this limitation, we propose utilizing auxiliary information from high-SNR bands as priors to guide the denoising of low-SNR bands. Our framework reformulates the optimization problem as follows:

$$\min_{\mathcal{X}_n} \|\mathcal{Y}_n - \mathcal{X}_n\|_1 + \lambda\phi\left(\mathcal{X}_n; \mathcal{X}_{\text{guide}}\right). \tag{3}$$

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Using the HySime algorithm, we estimate the noise intensity of $\mathcal{Y}$ and divide the spectral bands into $\mathcal{Y}_n$ and $\mathcal{X}_{\text{guide}}$ based on a predefined guidance ratio. Here, $\mathcal{Y}_n$ represents noisy low-SNR bands requiring denoising, and $\mathcal{X}_{\text{guide}}$ denotes high-SNR guiding bands. In this formulation, $\mathcal{X}_n$ is the denoised low-SNR bands to be recovered, and $\phi(\mathcal{X}_n; \mathcal{X}_{\text{guide}})$ integrates auxiliary information from $\mathcal{X}_{\text{guide}}$. Explicitly defining $\phi(.)$ to capture the complex spatial and spectral correlations between $\mathcal{X}_n$ and $\mathcal{X}_{\text{guide}}$ is challenging. To overcome this, we implicitly model the regularization term using a neural network G_θ, parameterized by θ

$$\mathcal{X}_n = G_\theta\left(\mathcal{Y}_n, \mathcal{X}_{\text{guide}}\right). \tag{4}$$

This reformulates the optimization problem as

$$\min_\theta \left\| \mathcal{Y}_n - G_\theta\left(\mathcal{Y}_n, \mathcal{X}_{\text{guide}}\right) \right\|_1. \tag{5}$$

Here, the regularization term is implicitly embedded within the network architecture, eliminating the need for explicit formulation. The network efficiently encodes spatial information from $\mathcal{X}_{\text{guide}}$, capturing multiscale spatial details crucial for preserving the structural integrity of the image. Our approach leverages the DIP mechanism, where the network architecture itself serves as an implicit regularizer. In DIP, even with a simple loss function like ours, the network tends to represent clean signals rather than noise patterns. The network can prevent overfitting to noise through appropriate early stopping strategies. Through guiding fusion modules (GFEM, UFEM) and the spatial–spectral attention unit (SSAU), the regularization is inherently integrated into the learning process, ensuring effective information transfer from high-SNR bands to low-SNR bands.

Our framework operates on the observed input $\mathcal{Y}$, utilizing noise estimation and manually defined guidance ratios to divide spectral bands. The network implicitly integrates multiscale spatial details and semantic features from the guiding bands, functioning as a dynamic regularizer to enhance denoising.

The network parameters θ are optimized by minimizing the loss function $L(\theta)$ using the Adam optimizer

$$L(\theta) = \left\| \mathcal{Y}_n - G_\theta\left(\mathcal{Y}_n, \mathcal{X}_{\text{guide}}\right) \right\|_1. \tag{6}$$

This iterative optimization balances noise suppression and detail preservation, which is critical for HSI denoising. By effectively integrating spatial and spectral information, the network serves as a dynamic regularizer, ensuring the preservation of essential features during the denoising process.

Our zero-shot denoising framework introduces a novel approach to HSI denoising, leveraging auxiliary high-SNR band information through a guided spectral–spatial fusion denoising network. The proposed method achieves an excellent balance between noise suppression and detail preservation, maintaining the integrity of spatial and spectral information in HSIs. The overall procedure is summarized in Algorithm 1.

C. Guided Spectral–Spatial Fusion Denoising Network

Training with paired noisy and clean HSIs is often costly and impractical, as clean data are unavailable. In contrast, unsupervised learning, particularly for complex tasks like HSI denoising, focuses on designing network architectures that are tailored to real-world application needs. Inspired by the inherent correlations among spectral bands and the availability of high-SNR bands in noisy hyperspectral data, we propose the guided spectral–spatial fusion denoising network.

Before the network iteration, we first estimate the noise level of the observed HSI $\mathcal{Y}$ using the HySime algorithm [43], as illustrated in Fig. 2. Let $E_i \in \mathbb{R}^N$ be the pixel column vector from the ith band, and form the complement matrix

$$E_{\partial i} = \left[E_1, \ldots, E_{i-1}, E_{i+1}, \ldots, E_L\right] \in \mathbb{R}^{N\times(L-1)}. \tag{7}$$

HySime performs a band-wise linear regression

$$E_i = E_{\partial i}\,\alpha_i + n_i \tag{8}$$

where α_i is computed by least squares, and the residual is

$$n_i = E_i - E_{\partial i}\,\alpha_i. \tag{9}$$

The residual n_i is treated as the noise sample for band i. Repeating this process for all L bands yields the residual matrix $[n_1, \ldots, n_L]$. The sample noise covariance is then given by

$$R_n = \frac{1}{N}\,[n_1, \ldots, n_L]^\top\,[n_1, \ldots, n_L]. \tag{10}$$

The diagonal elements of R_n represent the noise intensity of each band. Based on the guidance ratio, we automatically partition the spectral bands into high-SNR bands with lower noise and low-SNR bands with higher noise, thereby providing a reliable basis for denoising.

The proposed guided spectral–spatial fusion denoising network consists of three key components: AuxiSpectral Net, AdaptiveDenoising Net, and Guiding Fusion Modules.

The AuxiSpectral Net (guide branch) is designed to extract and refine multiscale spatial features from high-SNR guiding bands using an encoder–decoder architecture. In the encoder phase, convolutional layers with channel normalization progressively downsample the input, capturing hierarchical features at multiple scales. These features undergo nonlinear transformations to enhance the model's representation capacity. During the decoder phase, upsampling operations combined with skip connections preserve both high-level semantic and low-level detail information, ensuring comprehensive feature extraction from the guiding bands. The AdaptiveDenoising Net (denoising branch) mirrors the structure of the guide branch but focuses on noise suppression in low-SNR bands. It incorporates adaptive convolutional layers that adjust dynamically to the specific noise characteristics of each input band. The network's adaptability is further enhanced by two specialized fusion modules: GFEM for feature enhancement and UFEM for upsampling refinement. These modules enable the network to effectively leverage guiding information throughout the denoising process, ensuring improved noise reduction and detail preservation.

The Guiding Fusion Modules, consisting of the GFEM and the UFEM, include the SSAU. Attention mechanisms [51], particularly in image processing, have shown great promise in focusing on critical feature regions, improving the model's ability to capture essential information. The SSAU effectively

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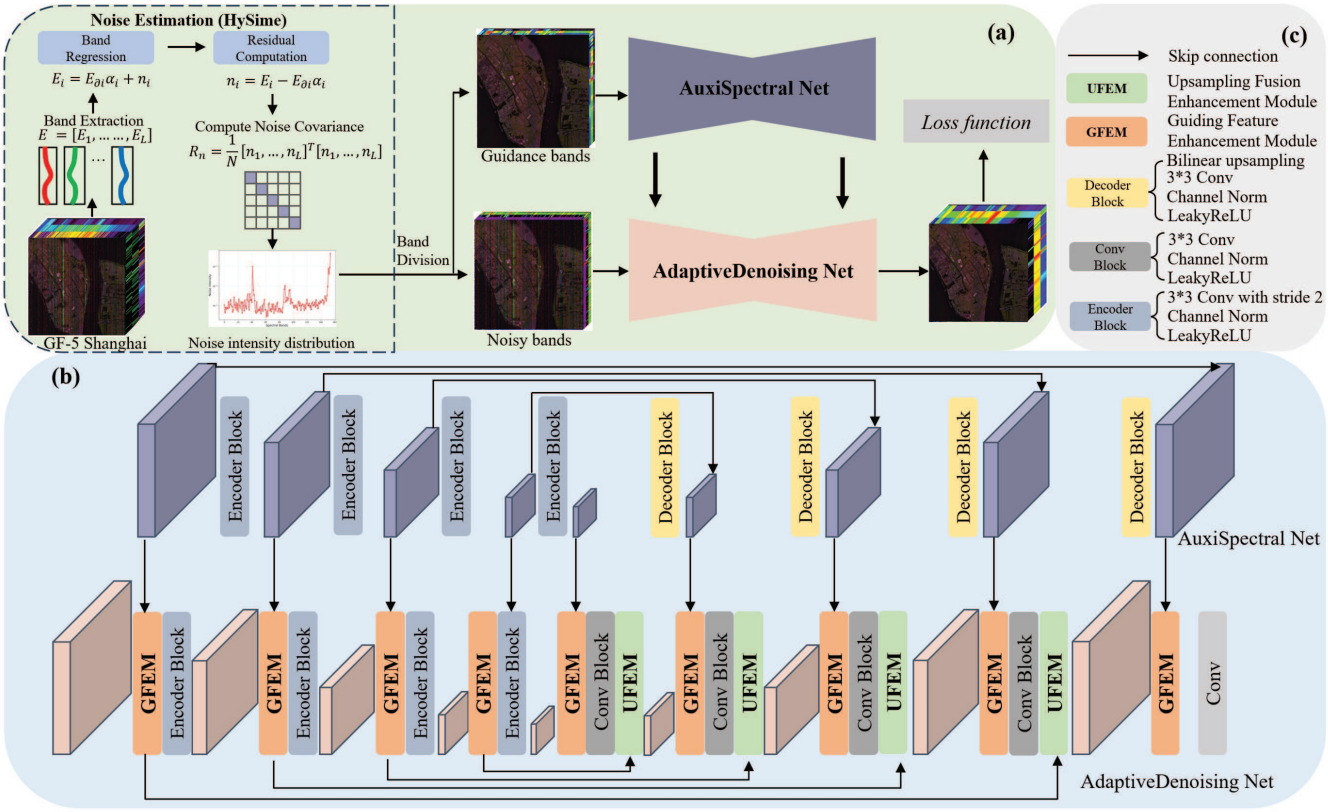


Fig. 2. Architecture of the zero-shot framework for HSI denoising. (a) Overall architecture of the zero-shot framework. (b) Detailed network structure. (c) Operations used in the network.

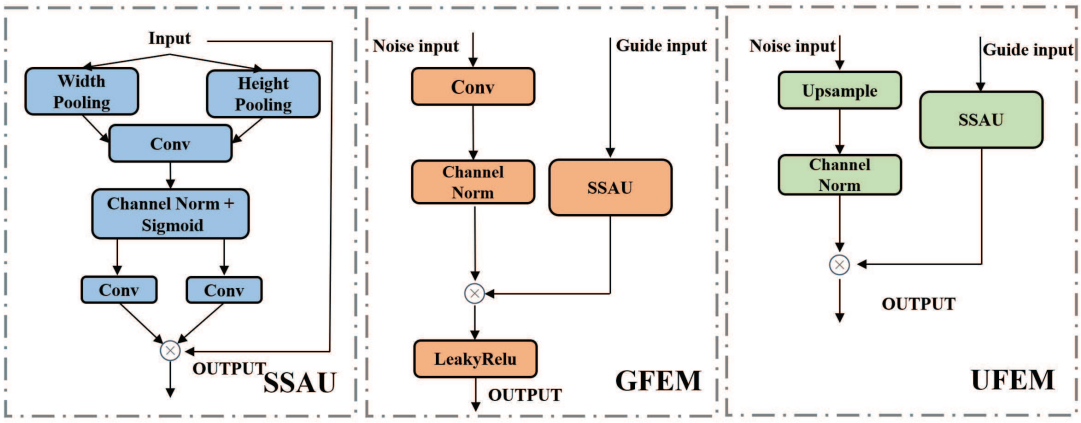


Fig. 3. Detailed architecture of the SSAU, GFEM, and UFEM modules.

combines both spectral and spatial information, enhancing the model's ability to distinguish between noise and valuable signal in HSIs. These modules are designed to process and fuse features from both high-SNR and low-SNR bands, significantly boosting denoising performance and improving overall image quality. Fig. 3 illustrates the detailed architecture of the SSAU, GFEM, and UFEM components within the network.

In SSAU, to effectively extract and integrate information from the guiding bands, we designed the SSAU to capture and utilize spatial structure information from the guiding bands, guiding the denoising process of noisy bands by dynamically adjusting the spatial sensitivity of features to optimize denoising performance. The SSAU begins by applying adaptive pooling to extract global spatial information from the image along both the height and width directions, generating feature maps Z_H and Z_W that capture the overall spatial structure of the guiding bands

$$Z_H = \text{AvgPool}_{H\times 1}\left(F_g\right), \quad Z_W = \text{AvgPool}_{1\times W}\left(F_g\right). \tag{11}$$

These feature maps are concatenated and processed through a shared 1×1 convolutional layer to fuse the spatial information

$$\delta = \text{Conv}_{1\times 1}\left([Z_H, Z_W]\right). \tag{12}$$

To generate accurate spatial attention maps, the fused features δ are split into two components, which are first processed

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through channel normalization and a Sigmoid activation function, and then further processed through independent 1×1 convolutions to produce the attention weights g_H and g_W

$$g_H = \mathrm{Conv}_{1\times1}(\delta), \quad g_W = \mathrm{Conv}_{1\times1}(\delta). \tag{13}$$

These attention weights are then applied to the original input feature map, dynamically adjusting the feature response at each pixel and producing the final attention-modulated output

$$F_{\mathrm{SSAU}} = F_g \times (g_H \otimes g_W) \tag{14}$$

where $\otimes$ denotes element-wise multiplication. The SSAU effectively suppresses noise while preserving spatial details, which is particularly important in high-noise environments typical of HSIs. This adaptive capability enables the SSAU to excel in HSI denoising, particularly in maintaining spatial structure information, which contributes to its superior performance.

The GFEM dynamically refines spatial and spectral features of low-SNR bands using attention mechanisms provided by the SSAU. GFEM generates attention maps that highlight crucial spatial and spectral information from the guiding bands, which are then applied to the low-SNR bands. This approach effectively reduces noise while preserving both spatial structures and spectral fidelity.

GFEM begins by extracting feature information from the guiding bands F_g and employs the SSAU to obtain spatial attention weights W_1

$$W_1 = \mathrm{SSAU}\left(F_g\right). \tag{15}$$

For the noisy band input F_n, GFEM processes the features using a 1×1 convolution layer, followed by channel normalization to ensure stability and adjust the feature distribution

$$F_n' = \mathrm{Channel\ Norm}\left(\mathrm{Conv}_{1\times1}(F_n)\right). \tag{16}$$

Next, GFEM applies the attention weights W_1 from the guiding bands to adjust the noisy band features in a weighted manner

$$F_{\mathrm{GFEM}} = F_n' \otimes W_1. \tag{17}$$

Finally, the adjusted features are activated by the LeakyReLU function to produce the optimized output features

$$F_{\mathrm{out}} = \mathrm{LeakyReLU}(F_{\mathrm{GFEM}}). \tag{18}$$

This process ensures that the spatially adaptive feature tuning, guided by the high-SNR bands, enhances the denoising performance while preserving essential spatial and spectral details.

The UFEM refines the features of low-SNR bands during the upsampling process, utilizing the SSAU to generate attention maps. These attention maps are applied during upsampling to preserve high-resolution spatial and spectral details from the guiding bands. By integrating the attention maps with the upsampled features, UFEM ensures that the denoised output retains both spatial structure and spectral fidelity.

UFEM begins by upsampling the noisy input F_n using bilinear interpolation, followed by channel normalization to stabilize the feature distribution

$$F_n^{\uparrow} = \mathrm{Channel\ Norm}\left(\mathrm{Upsample}(F_n)\right). \tag{19}$$

Simultaneously, the guiding bands F_g generate spatial attention weights W_2 through the SSAU

$$W_2 = \mathrm{SSAU}\left(F_g\right). \tag{20}$$

Algorithm 1 Zero-Shot Framework for Hyperspectral Image Denoising

Require: Observed HSI $\mathbf{Y}$, guidance ratio ω, maximum iteration T_{max}

Ensure: The final denoised HSI **out**

Initialize iteration $t \leftarrow 1$, network parameters θ
Stage I:
$\mathbf{N} \leftarrow \mathrm{HySime}(\mathbf{Y})$
$\{\mathbf{Y}_n, \mathbf{X}_{guide}\} \leftarrow \mathrm{Band_Division}(\mathbf{Y}, \mathbf{N}, \omega)$
Stage II:
while $t \le T_{max}$ **do**
  (a) Compute $\{\mathbf{F}_g, \mathbf{F}_n\}$ via Guided Spectral–Spatial Fusion Denoising Network:
  • $\mathbf{F}_g \leftarrow \mathrm{AuxiSpectral\ Net}(\mathbf{X}_{guide})$
  • $\mathbf{F}_n \leftarrow \mathrm{AdaptiveDenoising\ Net}(\mathbf{Y}_n)$
  (b) Fuse $\mathbf{F}_g$ and $\mathbf{F}_n$: $\mathbf{F}_{fused} \leftarrow \mathrm{GFEM}(\mathbf{F}_g, \mathbf{F}_n)$
  (c) Output the denoised result: $\mathbf{F}_{out} \leftarrow \mathrm{UFEM}(\mathbf{F}_{fused})$
  (d) Update network parameters $\theta(t-1) \leftarrow \theta(t)$ by optimizing:
$$L(\theta) = \|\mathbf{Y}_n - \mathbf{F}_{out}\|_1$$
  (e) Set $t \leftarrow t + 1$
end while

These weights are then applied to the upsampled features through element-wise multiplication

$$F_{\mathrm{UFEM}} = F_n^{\uparrow} \otimes W_2. \tag{21}$$

This operation enables spatially adaptive refinement driven by the guiding bands, effectively enhancing spatial precision and spectral consistency in the reconstructed output. By integrating the SSAU within both the GFEM and UFEM, the Guiding Fusion Module enables dynamic, spatially adaptive feature refinement guided by high-SNR bands. This approach effectively addresses the challenges posed by noise in hyperspectral imaging, significantly enhancing denoising performance and ensuring that the final output preserves both spectral accuracy and spatial detail, which are crucial for high-quality HSI reconstruction.

IV. Experimental Results and Analysis

In this section, we evaluate the effectiveness and efficiency of the proposed method through both synthetic and real-world experiments. We first introduce the compared methods, experimental setup, datasets, and evaluation metrics used for performance assessment.

We include the following representative state-of-the-art methods for comparison.

Low-Rank-Based Methods: The low-rank matrix recovery (LRMR) method [10], and the nonconvex fraction function (NFF) method [52].

Nonlocal Based Methods: The nonlocal meets global (NGmeet) method [53], and the tensor subspace representation denoising (TenSRDe) method [54].

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Total-Variation-Based Methods: The group sparsity regularized low-rank tensor decomposition (LRTDGS) method [30], and the E3DTV method [32].

Deep-Learning-Based Methods: The least favorable distribution-based deep image prior (HLF-DIP) method [41], the hierarchical low-rank tensor factorization (HLRTF) network [55], and the self-supervised CNN transform and TV (CTTV) regularization method [56].

For the synthetic experiments, we randomly selected a predetermined proportion of spectral bands from clean HSI datasets to form the guiding band set. The remaining bands were treated as noisy, to which various types of noise were added to simulate real-world degradation. In the real-world experiments, we used the HySime algorithm to estimate the noise level of each spectral band in the HSI. The bands were then sorted according to their predicted noise intensity: those with noise levels below a predefined threshold were designated as guiding bands, while the higher-noise bands were classified as noisy bands requiring denoising. We employed several evaluation metrics to assess performance, including peak SNR (PSNR), structural similarity index (SSIM), feature similarity index (FSIM), erreur relative globale adimensionnelle de synthèse (ERGAS), and spectral angle mapper (SAM). To ensure fair and consistent comparisons, we carefully tuned the parameters of all competing methods to their optimal settings, as recommended by the authors' code or their respective publications.

A. Synthetic Experiments

For the synthetic experiments, we utilized two widely recognized HSI datasets. The first dataset, the Washington DC Mall (WDC) [28], was acquired by the hyperspectral digital imagery collection experiment (HYDICE) sensor. It contains 210 spectral bands with a spatial resolution of 1208×307 pixels, covering wavelengths from 0.4 to 2.4 μm, with 191 noise-free bands. A clean subimage of size 288×288 pixels was chosen for testing. The second dataset, the Pavia Centre [28], was acquired by the reflective optics system imaging spectrometer (ROSIS-03). This dataset consists of 102 spectral bands with a spatial resolution of 1400×512 pixels, covering wavelengths from 430 to 860 nm. For analysis, we extracted a noise-free subimage of 192×192 pixels containing 80 spectral bands. In our experimental setup, 20% of the bands from the Pavia Centre dataset and 10% from the WDC dataset were randomly chosen to serve as guidance. To simulate real-world degradation, six different types of noise were added to the remaining spectral bands. These noise configurations were designed to represent various noise patterns and combinations, closely mimicking the complexities of real-world degradation. By employing this strategy, we comprehensively evaluated the performance of our algorithm in handling a wide range of challenging noise conditions.

1) *Case 1 (Independent and Identically Distributed Gaussian Noise):* The remaining bands of the HSI were corrupted by Gaussian noise with a standard deviation of 0.3.
2) *Case 2 (Nonindependent and Identically Distributed Gaussian Noise):* Each of the remaining bands was affected by Gaussian noise, where the standard deviation σ was randomly drawn from the interval [0.1, 0.3].
3) *Case 3 (Independent and Identically Distributed Impulse Noise):* The remaining bands were corrupted by impulse noise at a level of 30%.
4) *Case 4 (Nonindependent and Identically Distributed Impulse Noise):* Each band of the remaining HSI was corrupted by impulse noise, with the noise percentage randomly chosen from the range [0.1, 0.3].
5) *Case 5 (Gaussian Noise Combined With Impulse Noise):* Gaussian noise was first added to each remaining band, with σ randomly chosen from [0.1, 0.3]. Subsequently, impulse noise was introduced, with the noise percentage randomly selected from the range [0.1, 0.3].
6) *Case 6 (Mixture Noise):* Both Gaussian and impulse noises were applied, as described in Case 5. Additionally, 30 randomly selected bands were affected by deadlines, with the number of deadlines ranging from 3 to 12, and their widths randomly assigned between 1 and 3. Another set of 20 randomly chosen bands was affected by stripe noise, with the number of stripes fluctuating between 20 and 40.

Tables I and II present the quantitative evaluation results of all compared methods on the WDC and Pavia Centre datasets under various noise levels. The consistent results in both tables clearly indicate that our proposed method achieves the best or second-best performance across multiple quantitative metrics in HSI denoising tasks; in the tables, the best scores are shown in bold, while the second-best scores are underlined. It is noteworthy that under complex noise conditions, our method demonstrates significant advantages over existing methods. Cases 1 and 2 are mainly affected by Gaussian noise, for which existing methods are specifically designed and therefore already exhibit strong denoising performance. However, when Cases 3 to 6 involve sparse noise, methods based on NSS and patch-based modeling (e.g., TenSRDe and NGmeet) perform poorly. This is mainly because sparse noise disrupts both the structural similarity between patches and the low-rank characteristics within patches, negatively affecting the matching of similar patches and their clustering modeling, ultimately degrading denoising performance. To better handle the sparse, high-amplitude outliers typically present in sparse noise, we adopted the $L1$ loss (assuming the noise follows a Laplacian distribution) and introduced guiding bands to provide reliable structural information. These enhancements assist the network in more effectively distinguishing between image signals and noise. Thanks to these two design choices, our method achieves substantial PSNR improvements in Cases 3 and 4 and notable performance gains in Cases 5 and 6.

In Fig. 4, we present the pseudocolor results of noise removal from a mixed noise scenario (Case 6) on the Pavia Centre dataset, which includes four types of noise: Gaussian, impulse, deadlines, and stripes. The figure also includes corresponding error maps and zoomed-in views of marked regions for each comparative method. As shown in Fig. 4, methods such as CTTV and NGmeet fail to fully remove the noise, while methods like LRTDGS and E3DTV introduce

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TABLE I

QUANTITATIVE EVALUATION OF SIMULATED EXPERIMENTS WITH 20% HIGH-SNR BANDS FOR GUIDANCE ON PAVIA CENTRE DATASET

Case	Metrics	Low-Rank		Non-Local		Total-Variation		Deep Learning			Ours
		LRMR	NFF	NGmeet	TenSRDe	LRTDGS	E3DTV	HLF-DIP	HLRTF	CTTV	
1	PSNR	24.404	25.569	32.176	31.946	24.437	31.271	31.655	**35.738**	26.843	33.842
	SSIM	0.662	0.732	0.921	0.914	0.595	0.905	0.918	**0.959**	0.735	0.955
	FSIM	0.834	0.863	0.949	0.942	0.750	0.935	0.952	**0.976**	0.883	0.972
	ERGAS	220.364	193.231	89.981	92.391	218.388	100.540	95.246	**63.689**	232.188	74.998
	SAM	21.191	18.833	5.610	6.865	7.189	5.277	5.135	**4.568**	19.925	5.641
2	PSNR	27.621	32.595	29.573	33.740	25.249	32.828	32.648	**36.752**	31.267	34.761
	SSIM	0.764	0.937	0.831	0.939	0.649	0.932	0.929	**0.969**	0.830	0.959
	FSIM	0.880	0.958	0.908	0.959	0.784	0.955	0.960	**0.982**	0.916	0.976
	ERGAS	195.298	92.429	194.805	76.121	199.368	94.100	85.348	**54.130**	235.073	68.143
	SAM	19.400	7.211	18.318	6.050	7.024	5.945	4.461	**3.934**	21.553	5.037
3	PSNR	21.488	40.582	20.220	20.065	27.797	41.400	32.717	41.818	37.667	**47.194**
	SSIM	0.621	0.986	0.740	0.733	0.802	0.991	0.937	0.992	0.879	**0.997**
	FSIM	0.832	0.990	0.891	0.892	0.877	0.995	0.964	0.996	0.940	**0.998**
	ERGAS	320.105	44.347	373.640	377.837	149.385	38.183	85.459	30.789	245.695	**16.667**
	SAM	14.193	3.456	8.800	8.706	6.024	2.350	3.588	2.359	16.184	**1.759**
4	PSNR	25.785	41.461	22.489	23.671	27.961	42.899	32.290	43.423	37.309	**45.934**
	SSIM	0.754	0.988	0.694	0.824	0.812	0.993	0.930	0.993	0.863	**0.996**
	FSIM	0.882	0.992	0.860	0.927	0.884	0.996	0.960	0.996	0.930	**0.998**
	ERGAS	241.785	40.251	357.543	272.034	146.037	33.939	89.335	26.434	311.034	**18.994**
	SAM	14.586	3.189	14.971	9.613	6.167	2.448	3.780	2.284	24.277	**2.087**
5	PSNR	20.904	30.712	19.924	20.548	24.991	32.629	32.262	21.209	27.480	**35.124**
	SSIM	0.571	0.893	0.599	0.717	0.636	0.928	0.925	0.792	0.756	**0.965**
	FSIM	0.814	0.937	0.831	0.875	0.775	0.951	0.957	0.921	0.871	**0.979**
	ERGAS	358.980	144.587	434.127	368.563	207.935	98.227	90.489	393.194	342.874	**65.281**
	SAM	15.885	12.129	15.977	9.755	7.940	6.180	4.855	11.975	29.116	**4.654**
6	PSNR	21.466	30.186	20.607	21.311	24.617	32.487	31.466	22.124	28.246	**33.369**
	SSIM	0.589	0.888	0.623	0.733	0.625	0.930	0.919	0.817	0.779	**0.953**
	FSIM	0.818	0.936	0.838	0.881	0.772	0.951	0.954	0.931	0.880	**0.973**
	ERGAS	342.499	159.077	410.286	343.920	216.791	96.045	99.326	362.240	350.040	**80.523**
	SAM	16.222	13.825	16.507	10.003	8.934	**6.110**	6.781	11.929	29.377	6.334

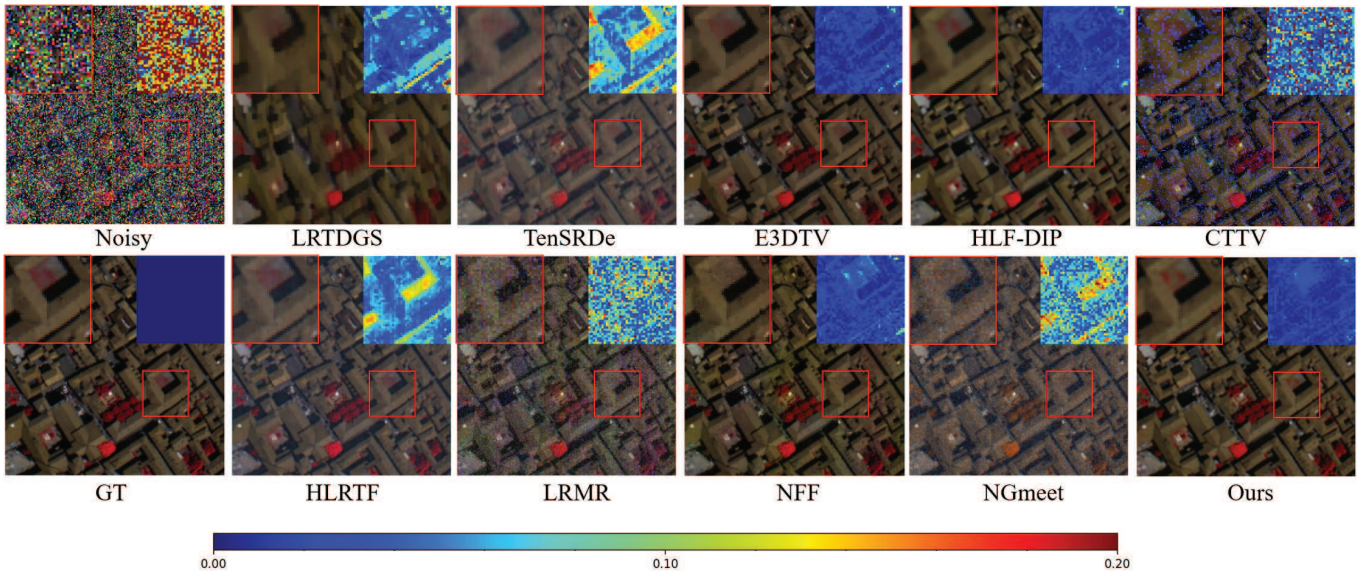


Fig. 4. Mixture noise removal results in Case 6, visualized by pseudocolor images on the Pavia Centre dataset.

excessive smoothing, resulting in the loss of fine details in the original image. In contrast, our approach maintains spectral details while retaining the integrity of spatial structures. Figs. 5 and 6 present the pseudocolor denoising results and corresponding error maps for mixed noise scenarios (Cases 5 and 6) on the WDC dataset, along with zoomed-in views of marked regions for each comparative method. As shown in these figures, our method consistently delivers the best results across a wider range of spectral bands compared to other methods.

These results highlight the superior performance of our approach, particularly in handling complex noise scenarios while ensuring both spatial and spectral integrity. Fig. 7 shows the spectral signature curves at pixel location (80, 150)

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TABLE II
QUANTITATIVE EVALUATION OF SIMULATED EXPERIMENTS WITH 10% HIGH-SNR BANDS FOR GUIDANCE ON WDC DATASET

Case	Metrics	Low-Rank		Non-Local		Total-Variation		Deep Learning			Ours
		LRMR	NFF	NGmeet	TenSRDe	LRTDGS	E3DTV	HLF-DIP	HLRTF	CTTV	
1	PSNR	26.824	15.927	**32.652**	32.582	26.493	28.823	25.871	25.031	31.676	32.002
	SSIM	0.754	0.250	**0.922**	0.919	0.682	0.829	0.791	0.747	0.910	0.914
	FSIM	0.883	0.611	**0.955**	0.952	0.796	0.886	0.893	0.877	0.945	0.945
	ERGAS	170.159	622.147	**88.326**	89.059	182.017	148.375	248.881	273.220	96.742	96.141
	SAM	11.866	35.792	**4.368**	4.967	7.830	8.318	17.304	19.930	5.214	5.394
2	PSNR	29.532	28.786	32.148	34.271	27.536	29.783	29.661	**35.867**	34.295	34.842
	SSIM	0.837	0.797	0.901	0.942	0.734	0.851	0.838	0.942	0.939	**0.956**
	FSIM	0.919	0.903	0.947	0.966	0.829	0.901	0.909	0.969	0.966	**0.972**
	ERGAS	132.775	162.013	112.107	72.829	159.945	127.771	124.793	**68.094**	73.269	68.427
	SAM	8.990	10.644	6.337	4.083	6.829	6.673	5.310	3.933	4.086	**3.660**
3	PSNR	22.871	38.634	20.168	20.308	30.971	37.854	30.380	45.280	**49.329**	46.260
	SSIM	0.690	0.974	0.667	0.704	0.873	0.979	0.869	0.995	**0.997**	**0.997**
	FSIM	0.846	0.985	0.888	0.894	0.922	0.989	0.927	0.939	**0.998**	**0.998**
	ERGAS	340.550	64.566	441.240	434.605	110.513	57.939	116.744	**16.369**	245.695	21.903
	SAM	20.736	3.374	16.425	16.413	4.517	2.999	4.553	18.94	**0.944**	1.262
4	PSNR	27.440	43.828	23.537	23.667	31.575	38.993	30.687	44.957	**49.751**	45.921
	SSIM	0.828	0.992	0.784	0.807	0.889	0.984	0.877	0.996	**0.997**	0.996
	FSIM	0.875	0.995	0.924	0.934	0.933	0.992	0.932	0.997	**0.998**	0.997
	ERGAS	308.876	28.019	315.802	299.846	102.420	49.868	111.569	21.878	**15.432**	22.688
	SAM	15.344	1.517	13.310	12.760	4.165	2.610	4.270	1.094	**0.892**	1.290
5	PSNR	21.707	22.438	20.822	20.891	26.549	29.256	29.610	21.972	32.973	**34.213**
	SSIM	0.641	0.561	0.684	0.698	0.703	0.847	0.846	0.792	0.923	**0.953**
	FSIM	0.858	0.797	0.889	0.884	0.810	0.900	0.916	0.925	0.957	**0.971**
	ERGAS	371.141	390.394	438.071	432.655	181.433	141.883	128.166	467.615	87.748	**78.871**
	SAM	17.745	23.722	17.507	17.230	8.158	7.797	5.810	18.737	5.094	**4.342**
6	PSNR	22.008	22.413	21.419	21.482	26.433	28.996	28.889	28.782	**32.457**	32.273
	SSIM	0.651	0.569	0.720	0.729	0.698	0.839	0.826	0.796	0.918	**0.919**
	FSIM	0.859	0.796	0.890	0.899	0.807	0.893	0.903	0.895	0.954	**0.959**
	ERGAS	361.185	388.988	415.386	412.435	180.455	141.712	136.226	165.421	**92.266**	95.340
	SAM	17.209	22.719	16.613	16.516	7.772	7.337	6.028	10.412	5.272	**5.019**

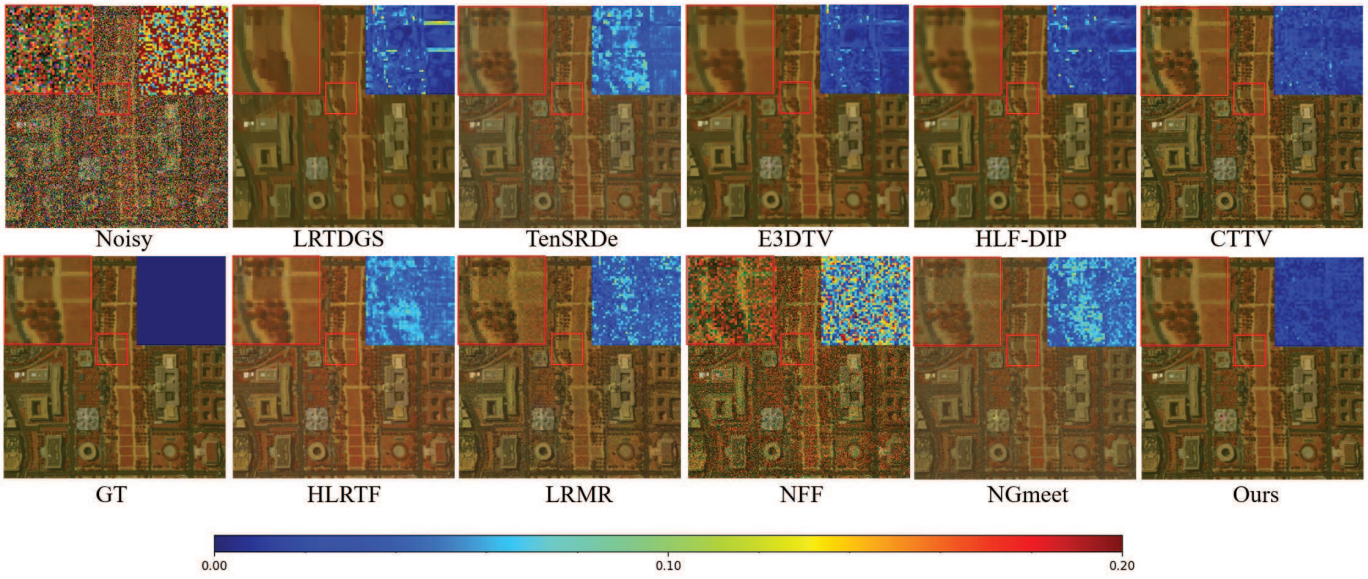


Fig. 5. Mixture noise removal results in Case 5, visualized by pseudocolor images on the WDC dataset.

on the WDC dataset under challenging conditions (Case 6). This further demonstrates the ability of our method to effectively remove noise while preserving spectral integrity across all bands. These findings emphasize the exceptional capability of our method in simultaneously achieving noise reduction and preserving spectral information.

B. Real-World Experiments

This section demonstrates the performance advantages of our method using real-world hyperspectral data. We evaluated our method using two real-world HSI datasets: GF-5 Shanghai and GRANHHADA [57]. The GRANHHADA dataset consists of 35 outdoor scenes captured under various atmospheric

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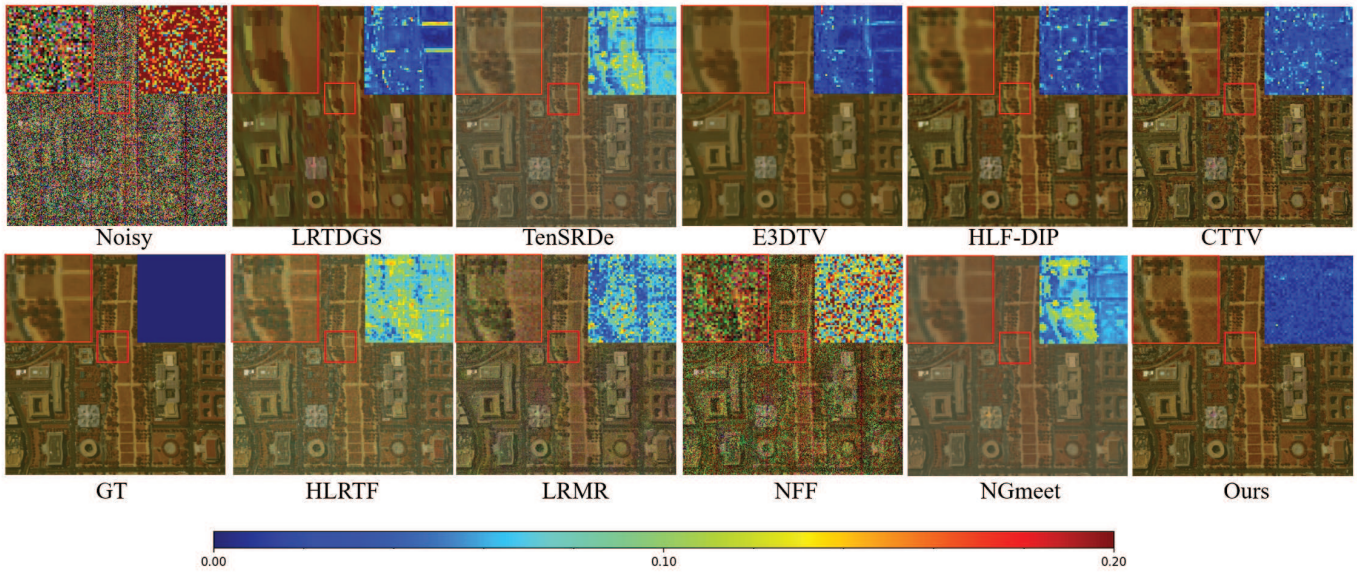


Fig. 6. Mixture noise removal results in Case 6, visualized by pseudocolor images on the WDC dataset.

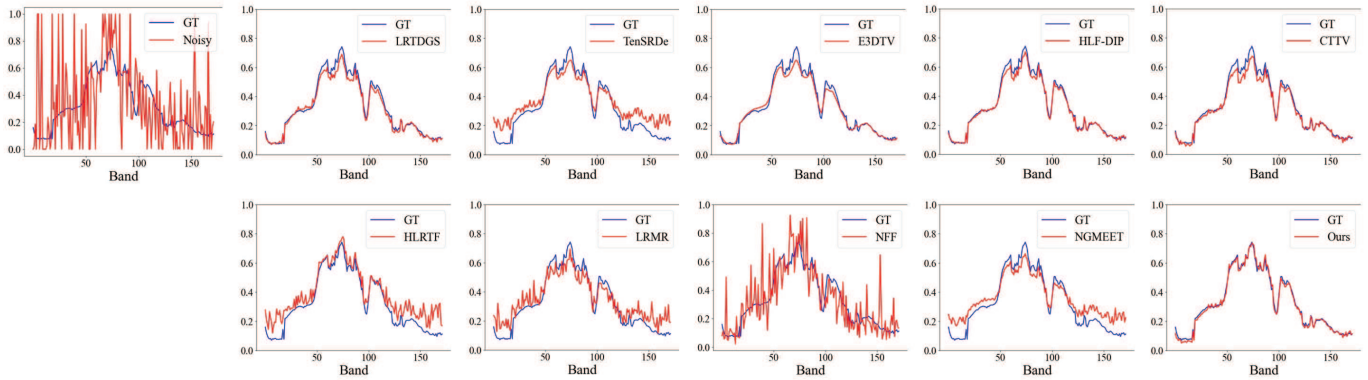


Fig. 7. Spectral signatures in (80, 150) of WDC dataset in Case 6.

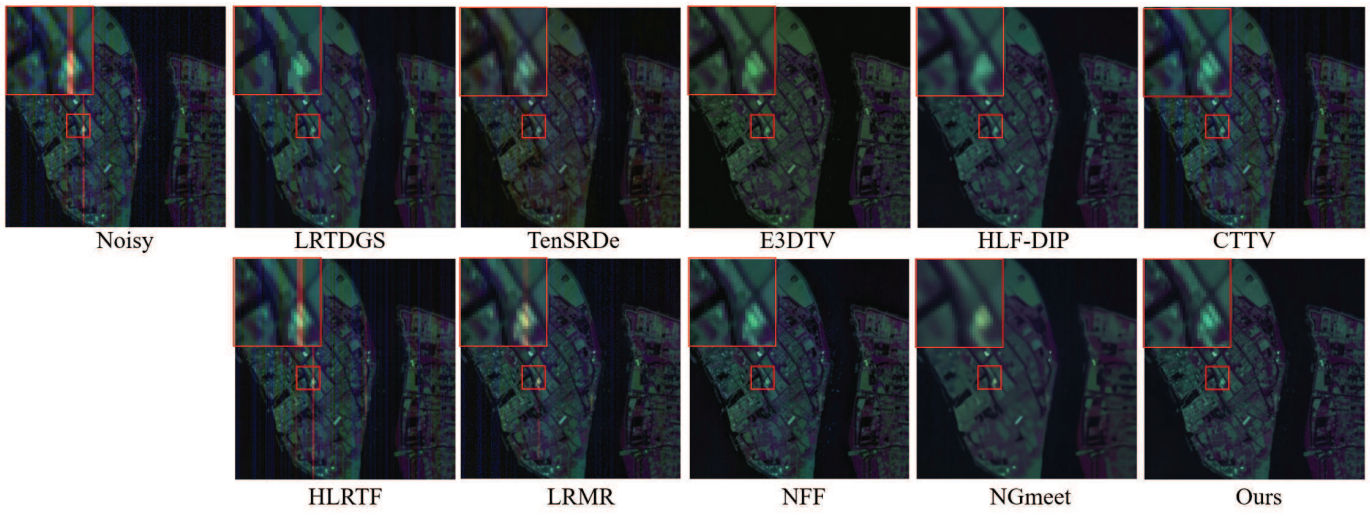


Fig. 8. Noise removal results visualized by pseudocolor images constructed from multiple bands of the GF-5 dataset.

conditions, including images from two hyperspectral cameras: the SPECIM-IQ, covering the visible/near-infrared (VIS/NIR) range with a resolution of 512×512 pixels and 204 spectral channels, and the PIKA NIR-640, capturing the shortwave infrared (SWIR) range with 168 merged spectral channels. For our experiments, we selected two images from this dataset. The GF-5 Shanghai dataset, captured by the GaoFen-5 satellite's Advanced Hyperspectral Imager, consists of HSIs

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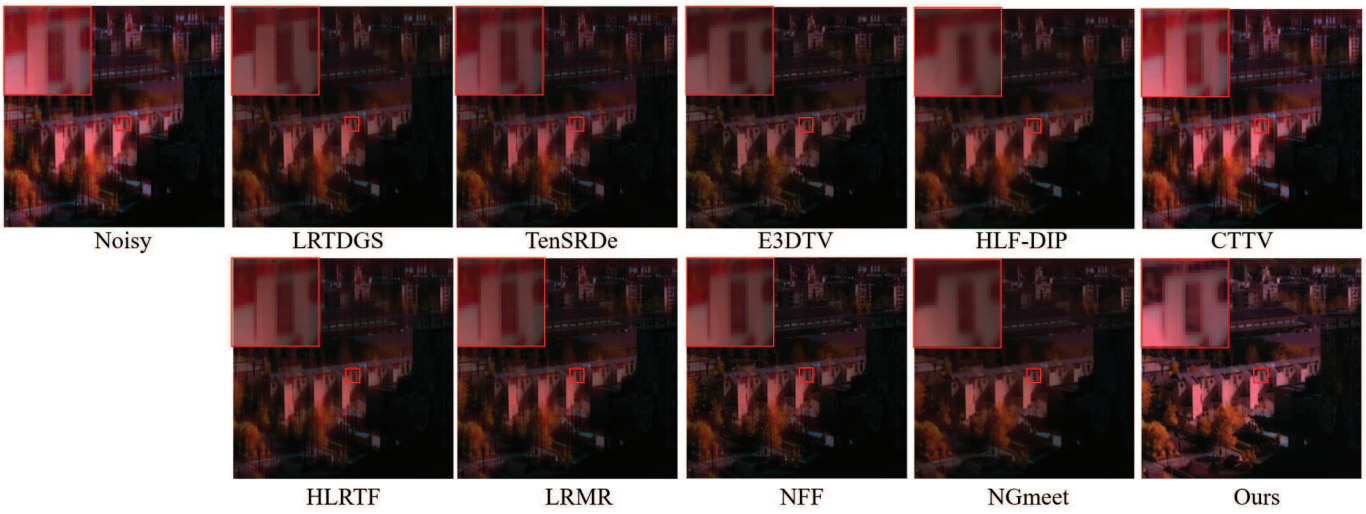


Fig. 9. Noise removal results visualized by pseudocolor images constructed from multiple bands Image-1 in the real Granhhada dataset.

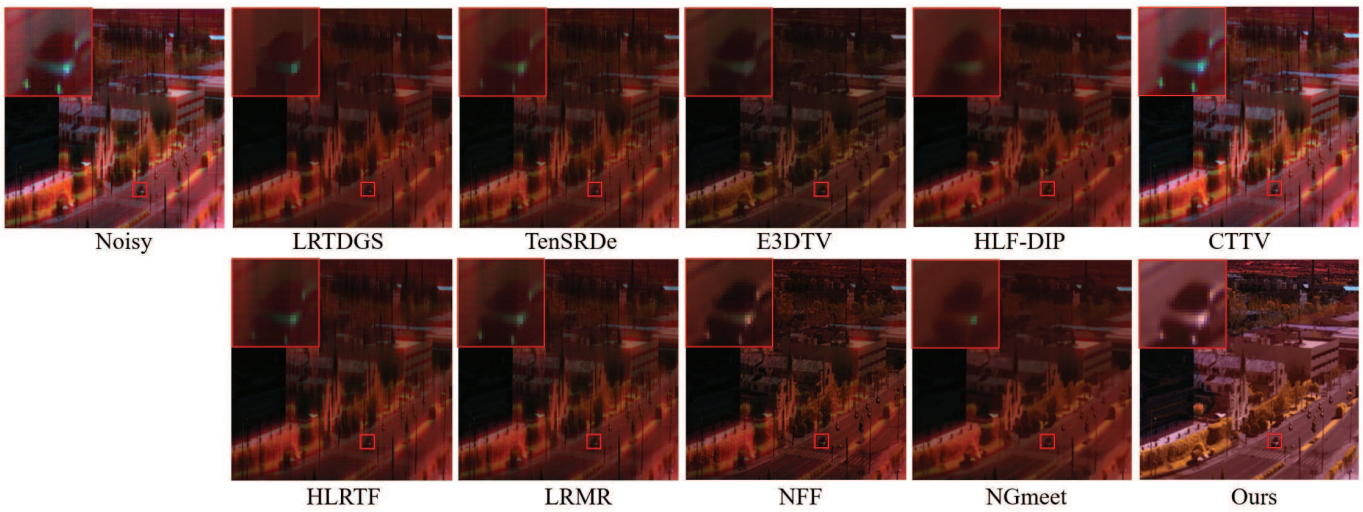


Fig. 10. Noise removal results visualized by pseudocolor images constructed from multiple bands of Image-2 in the real Granhhada dataset.

with a spatial resolution of 30 m per pixel and 330 continuous spectral bands spanning the 0.4–2.5 μm range. We selected a sub-image with dimensions of $288 \times 288 \times 155$ for our experiments. To extract high-SNR bands, we applied the HySime method to estimate the noise levels for each band and selected the top 10% of the bands with the lowest noise to serve as guiding bands. Bands affected by water and atmospheric absorption were excluded for improved analysis. Since ground truth is unavailable for real-world datasets, we cannot apply objective evaluation metrics. Instead, we conduct qualitative comparisons by generating pseudocolor images using multiple spectral bands and zoomed-in views. By visually comparing the denoised results, we can evaluate the effectiveness of different denoising methods.

Fig. 8 presents the noise removal results for the Shanghai dataset, which is heavily affected by prominent stripe noise. Our method delivers the most effective denoising performance compared to existing approaches. Specifically, methods such as CTTV fail to fully eliminate the visible stripe noise, while techniques like NGmeet manage to reduce the noise but at the cost of introducing excessive smoothing, leading to the loss of important image details. In contrast, our approach not only removes the severe stripe noise entirely but also preserves spatial details effectively. This demonstrates the reliability and superior performance of our method in striking a balance between noise reduction and the preservation of fine image features, highlighting its effectiveness for practical applications.

The denoising results on the GRANHHADA dataset are presented in Figs. 9 and 10. As illustrated, the original images are severely contaminated by substantial noise. In comparison to our method, most existing techniques fail to eliminate the noise completely, resulting in denoised images that remain heavily degraded. Notably, only the NGmeet method shows some degree of restoration. In contrast, our method achieves the best restoration results, effectively removing the pervasive noise while preserving clear spatial details without introducing artifacts or distortions. This remarkable performance highlights the robustness and efficacy of our method, which significantly outperforms other techniques in both noise reduction and detail preservation, showing its potential for high-quality image processing applications.

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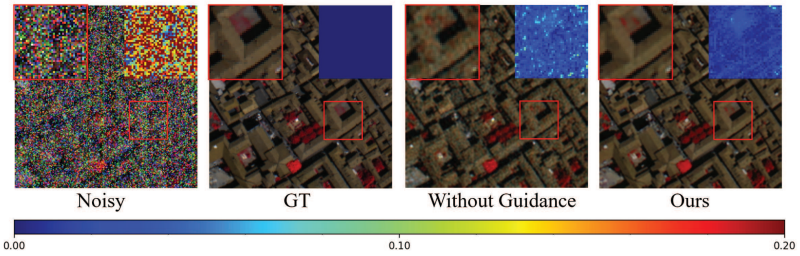


Fig. 11. Noise removal results visualized by the pseudocolor images for the Pavia Centre dataset.

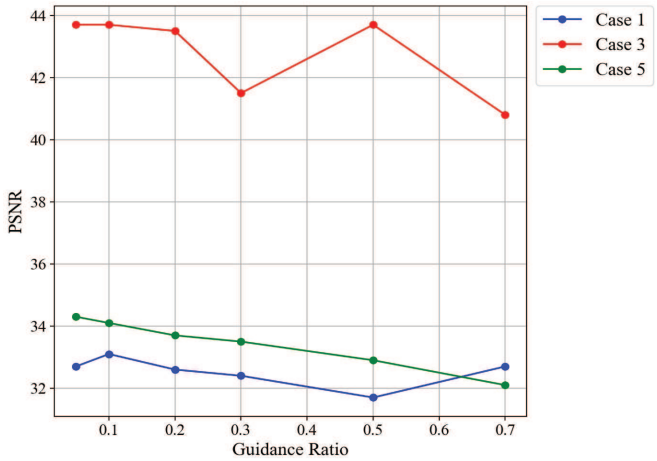


Fig. 12. Impact of guidance ratio (ω) on PSNR performance across different noise types on the Pavia Centre dataset.

TABLE III
BAND INDEX AND PSNR PERFORMANCE FOR DIFFERENT NOISE ESTIMATION METHODS

Method	Common Band Index	Different Band Index	PSNR
HySime	1, 4, 13, 21, 22, 33, 53, 58, 63, 64, 65, 75, 76	43, 50, 72	**33.37**
Wavelet-based	1, 4, 13, 21, 22, 33, 53, 58, 63, 64, 65, 75, 76	35, 48, 73	32.66
Eigenvalue-based	1, 4, 13, 21, 22, 43, 50, 58, 63, 64, 65, 75, 76	38, 54, 73	33.11

C. Ablation Study

In this article, we proposed a framework that integrates guiding spectral band information with a guided spectral–spatial fusion denoising network. To comprehensively evaluate our method, we conducted a series of ablation experiments.

Specifically, we employed three different noise estimation methods to estimate noise levels on the Pavia Centre dataset under complex noise conditions (Case 6): HySime, a wavelet transform-based estimation method [58], and an estimation method based on the eigenvalue distribution of image patches [59]. Despite their distinct theoretical foundations, the three methods showed high consistency when estimating noise from the same hyperspectral dataset, with only slight differences in a few spectral bands. As shown in Table III, denoising based on these estimated bands has little impact on the final results. This observation confirms that noise estimation serves merely as a necessary preprocessing step in our algorithm pipeline rather than a core component. Regardless of the specific noise estimation technique employed, the key requirement is to accurately select high-quality spectral bands. Consequently, our proposed method is insensitive to the particular choice of noise estimation technique.

TABLE IV
QUANTITATIVE RESULTS OF ABLATION STUDY ON DIFFERENT MODULES

Modules	PSNR	SSIM	SAM
w/o GFEM	29.98	0.8915	8.72
w/o UFEM	32.17	0.9172	6.13
w/o SSAU	32.79	0.9284	5.47
Full Model	**34.21**	**0.9530**	**4.34**

TABLE V
DENOISING PERFORMANCE WITH AND WITHOUT GUIDING BAND INFORMATION

Metrics	PSNR	SSIM	FSIM	ERGAS	SAM
Ours	**33.3686**	**0.9528**	**0.9732**	**80.5230**	**6.3343**
Without Guidance	28.6291	0.8343	0.8987	136.3644	8.2643

To evaluate the effectiveness of each component in the proposed model, we conducted ablation experiments under complex noise conditions (Case 5) using the WDC dataset, and the results are summarized in Table IV. Specifically, "w/o SSAU" denotes the removal of the SSAU, "w/o GFEM" refers to the removal of the GFEM, and "w/o UFEM" denotes the removal of the UFEM.

Experimental results show that replacing GFEM with a standard convolution and normalization structure leads to a performance drop of 4.23 dB in PSNR, indicating that the guided features significantly contribute to spectral–spatial detail preservation during reconstruction. When UFEM is substituted with bilinear interpolation, the PSNR drops by 2.04 dB, which validates the role of guided upsampling in maintaining fine details. Additionally, removing SSAU and replacing it with conventional convolutional layers and activation functions results in a PSNR decrease of 1.42 dB, highlighting the importance of spatial information learning in the fusion process.

We additionally perform a comprehensive analysis of guiding band information on denoising performance by comparing the denoising results from simulated experiments, both with and without the inclusion of this information. When the guidance from high-SNR bands and their associated spatial

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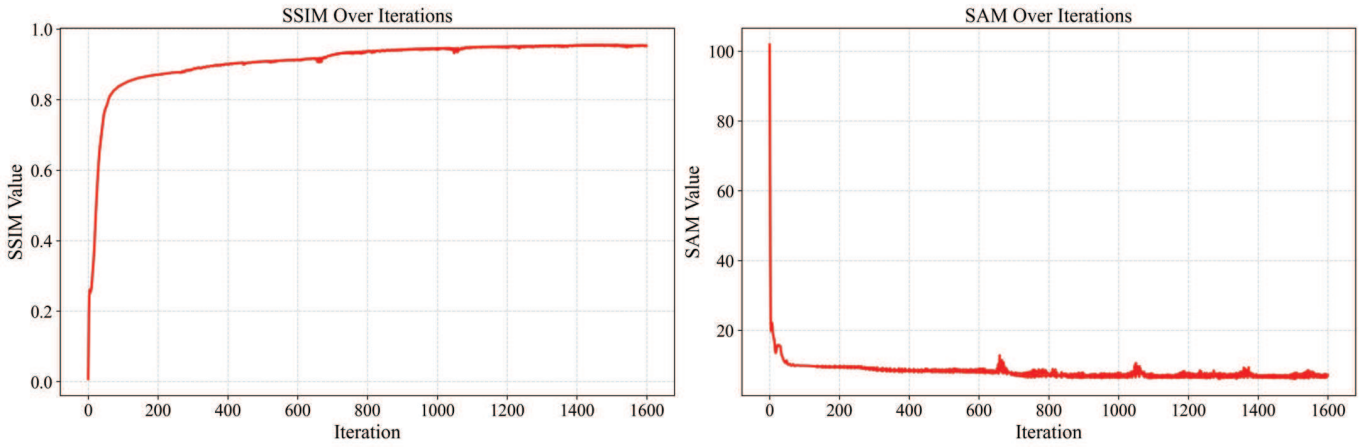


Fig. 13. Convergence analysis on the Pavia Centre dataset: evolution of SAM and SSIM values over iterations.

structures is removed, our proposed denoising network degenerates into a single-branch network that only processes noisy band inputs. The denoising results are visualized in Fig. 11, and the corresponding quantitative metrics are presented in Table V. Both the visualizations and the quantitative results demonstrate the critical role that guiding band information plays in enhancing the performance of our method.

To comprehensively analyze the impact of varying guidance ratios (ω) on denoising performance, we present the relationship between PSNR and the guidance ratio ω in Fig. 12, based on the Pavia Centre dataset. Across different noise types—whether Gaussian noise, sparse noise, or mixed noise—the PSNR curve shows only minor fluctuations within the range of [0.05, 0.7]. This indicates that changes in the guidance ratio have a relatively moderate effect on the overall denoising performance.

Further analysis reveals that while increasing the guidance ratio leads to slight performance gains, these improvements are not particularly pronounced. This suggests that the network remains effective at extracting essential information for denoising, even with lower levels of supervision, maintaining stability across various noise scenarios. These findings highlight the robustness of the proposed method, which delivers consistently high-quality denoising with minimal sensitivity to the guidance ratio.

Fig. 13 presents the convergence analysis on the Pavia Centre dataset, illustrating the evolution of SAM and SSIM values over iterations. The clear trend of stabilization as the number of iterations increases confirms the successful convergence of our iterative optimization algorithm, performed under uniform conditions.

V. Conclusion

In this article, we present a zero-shot HSI denoising framework that utilizes high-SNR bands to guide the denoising of low-SNR bands. By adopting a dual-branch architecture network and incorporating the SSAU in the feature fusion modules, our method significantly improves denoising performance without requiring paired training data. Extensive experiments demonstrate that our approach outperforms existing methods under various noise conditions, effectively preserving both spectral and spatial details. Future work will explore extending this framework to other HSI processing tasks, further enhancing its applicability across diverse domains.

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